

Java Order of Operations

Priority	Operator(s)	Description
1	()	Parentheses
2	**, Math.pow()	Exponentiation
3	!	Logical NOT
4	*, /, %	Multiplication, Division, Modulus
5	+, -	Addition, Subtraction
6	>, <, >=, <=	Relational / Comparison
7	==, !=	Equality
8	&&	Logical AND
9		Logical OR
10	=, +=, -=	Assignment

Java Control Flow

if

```
if (condition) {  
    // code block  
}
```

if/else

```
if (condition) {  
    // code block  
} else {  
    // code block  
}
```

Ternary if/else

```
variable = (condition) ? valueIfTrue : valueIfFalse;
```

if/else-if

```
if (condition1) {  
    // code block  
} else if (condition2) {  
    // code block  
} else {  
    // code block  
}
```

Switch

```
switch(expression) {  
    case x:  
        // code block  
        break;  
    case y:  
        // code block  
        break;  
    default:  
        // code block  
}
```

for loop

```
for (int i = 0; i < 5; i++) {  
    // code block  
}
```

while loop

```
while (condition) {  
    // code block  
}
```

do-while loop

```
do {  
    // code block  
} while (condition);
```

Java Methods

```
public static <returnType> <methodName>(<parameters>) {  
    // code that runs when the method is called  
    return <someValue> // Note: Don't need 'return' if <returnType> is `void`  
}
```

Java Arrays

Create with values

```
int[] numbers = {10, 20, 30, 40, 50};
String[] colors = {"red", "green", "blue"};
```

Create empty (fixed size)

```
int[] numbers = new int[5];
```

Access / Modify

```
numbers[0] // first element
numbers[2] = 99; // change third element
numbers.length // number of elements
```

for-each loop

```
for (int num : numbers) {
    // code block
}
```

Multidimensional Arrays (2D)

Create with values

```
int[][] grid = {
    {1, 2, 3},
    {4, 5, 6}
};
```

Access / Modify

```
grid[0][2] // row 0, column 2
grid[1][0] = 99; // change row 1, column 0
grid.length // number of rows
grid[0].length // number of columns
```

Create empty (fixed size)

```
int[][] grid = new int[2][3]; // 2 rows, 3 columns
```

Nested loop

```
for (int r = 0; r < grid.length; r++) {
    for (int c = 0; c < grid[r].length; c++) {
        System.out.print(grid[r][c] + " ");
    }
    System.out.println();
}
```

Nested for-each loop

```
for (int[] row : grid) {
    for (int val : row) {
        System.out.print(val + " ");
    }
    System.out.println();
}
```

ArrayList

Create

```
ArrayList<String> names = new ArrayList<String>();
```

Note: Must use object types (String, Integer, Double), not primitives.

Common methods

```
names.add("Alice"); // add to end
names.get(0); // access by index
names.set(0, "Bob"); // change element
names.remove("Alice"); // remove by value
names.size(); // number of elements
names.contains("Bob"); // check if exists
```

String Methods

```
s.equals("hi") // true if same text
s.equalsIgnoreCase("Hi") // ignores case
s.length() // number of characters
s.isEmpty() // true if ""
s.contains("fox") // true if found
s.toUpperCase() // "Hello" -> "HELLO"
s.toLowerCase() // "Hello" -> "hello"
```

```
s.trim() // remove leading/trailing spaces
s.startsWith("The") // true if starts with "The"
s.endsWith("dog") // true if ends with "dog"
s.replace("a", "b") // replace all "a" with "b"

// "a,b,c" -> {"a", "b", "c"}
String[] parts = s.split(",");
```